

1. (C) 4794 Through complementary counting, we can find the number of positive integers that divide 99 and subtract that from the sum of the first 99 positive integers. Doing so, we get

$$99 \cdot 50 - (1 + 3 + 9)(1 + 11) = 4950 - 156 = 4794$$

2. (B) 416 32_b is equal to $3b + 2$ in base 10. Thus, the summation is equivalent to $\sum_{b=4}^{16} 3b + 2$, which computes to 416.

3. (A) -4095 The best way to solve this is through complementary counting. The sum of the coefficients of $(4w - 3x + y - 3z)^{12} = (4 - 3 + 1 - 3)^{12} = 1$. The sum of the coefficients without an x term is $(4 - 3(0) + 1 - 3)^{12} = 4096$. $1 - 4096 = -4095$.

4. (B) 305 Under the conditions $a + b + c < 22$, $0 \leq a, b, c < 14$, and $a + b + c \equiv 3 \pmod{5}$, $a + b + c \in \{3, 8, 13, 18\}$. Using stars-and-bars, we get that there are $\binom{5}{2} + \binom{10}{2} + \binom{15}{2} + \binom{20}{2} = 350$ ordered triplets such that $a + b + c \in \{3, 8, 13, 18\}$. However, we must account for the cases where either a, b , or c are greater than or equal to 14. Because 3, 8, and 13 are all less than 14, we do not have to worry about extraneous cases. However, there are some cases for which $a + b + c = 18$ and one of the variables is greater than or equal to 14. To count the number of these cases, we choose one variable to equal at least 14 and perform stars-and-bars. Thus, the number of cases in which $\binom{3}{1} \cdot \binom{6}{2} = 45$. Thus, the number of cases that fulfill all of the problem's restrictions is $350 - 45 = 305$.

5. (E) NOTA Plugging in some values for the sequence, we see that $F_{-1} = 1, F_{-2} = -1, F_{-3} = 2, F_{-4} = -3, \dots$. From this, one can see that $F_{-n} = -F_n$ when n is even. Thus, $F_{2024} + F_{-2024} = 0 \pmod{6}$.

6. (C) 90 When two limits, $A = \lim_{x \rightarrow k} a(x)$ and $B = \lim_{x \rightarrow k} b(x)$, are defined and non-zero, then $\lim_{x \rightarrow k} a(x)b(x) = AB$. We can split this limit into $\lim_{x \rightarrow 0} \frac{1 - \cos(3x)}{x^2} \cdot \frac{\sin(5x)}{x} \cdot \frac{\tan(4x)}{x}$. Taking each limit individually through L'Hôpital's, we get that the final limit is equal to $\frac{9}{2} \cdot 5 \cdot 4 = 90$

7. (D) $\frac{31}{12}$ For a hyperbola of the form $f(x, y)g(x, y) = c$ where $c \neq 0$, there is no solution when either $f(x, y) = 0$ or $g(x, y) = 0$ and thus, are the asymptotes of the hyperbola. It is also known that the asymptotes of a hyperbola intersect at its center. Thus, the asymptotes are

$$\begin{cases} x - 4y + 2 = 0 \\ 2x + 3y - 7 = 0 \end{cases}$$

Solving the system for the center and adding the slopes of the asymptotes and the abscissa and ordinate of the center, we get $2 + 1 + \frac{1}{4} - \frac{2}{3} = \frac{31}{12}$

8. (C) 6070 Through sum-to-product,

$$\cos 4047x + \cos x = 2 \cos 2024x \cos 2023x.$$

Thus,

$$\cos 4047x + \cos x - 2 \cos 2023x = 2 \cos 2023x(\cos 2024x - 1) = 0.$$

From $x \in [0, 2\pi)$, there are 4046 solutions to $\cos 2023x = 0$ while there are 2024 solutions to $\cos 2024x = 0$. It is also quite easy to see that none of these solutions overlap. Thus, there are $4046 + 2024 = 6070$ solutions.

9. (D) $\frac{1}{8}$ The roots of this polynomial are 0 and the 16th roots of unity not including 1. $|r_2 - r_1| = 1$ only when either r_1 or r_2 is 0. There are $\binom{15}{1} = 15$ ways to do this out of a total of $\binom{16}{2} = 120$ possible values. Thus, the probability is $\frac{15}{120} = \frac{1}{8}$

10. (A) $\begin{bmatrix} e^2 & 0 \\ 2e^2 - 2e & e \end{bmatrix}$ It's obvious that the diagonals of the summation will be e^2 and e respectively. Additionally, it's also easy to see that the top right element will be 0; thus, what we need to solve is the bottom left element. Writing the first couple elements of the Taylor Series, we get $0, 2, 6, 14, 30, \dots$, which we can rewrite as $(2 - 2), (4 - 2), (8 - 2), (16 - 2), (32 - 2), \dots$. Taking these two sums separately and subtracting, we get that the bottom left element is $2e^2 - 2e$. Thus, e^M is

$$\begin{bmatrix} e^2 & 0 \\ 2e^2 - 2e & e \end{bmatrix}$$

Note that you can also diagonalize this matrix into PDP^{-1} and taking the summation inside of D . Doing this, you would get $P \begin{bmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{bmatrix} P^{-1}$ as your answer, which comes out to the same matrix.

11. (D) 48 By Extended Law of Sines, $\frac{a}{\sin A} = \frac{c}{\sin C} = 2R$. Through basic geometry formulas, $2R = \frac{abc}{2[ABC]}$. Thus, $\frac{a}{\sin A} = \frac{c}{\sin C} = \frac{ac}{24}$. Dividing all sides by a and b , we get $\frac{1}{c \sin A} = \frac{1}{a \sin C} = \frac{1}{24}$, meaning that $a \sin C = c \sin A = 24$; thus, $a \sin C + c \sin A = 48$.

12. (A) $\begin{bmatrix} 6 & 4 \\ 2 & 8 \end{bmatrix}$ Through basic left and right matrix multiplication, we can see that the diagonalization of

$$\begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{4} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}^{-1}$$

We can then substitute that matrix with its diagonalization, giving us

$$\sum_{n=0}^{\infty} \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}^n = \sum_{n=0}^{\infty} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{4} \end{bmatrix}^n \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}^{-1} \Rightarrow \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \left(\sum_{n=0}^{\infty} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{4} \end{bmatrix}^n \right) \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}^{-1}$$

We can now very easily take the sum of this diagonal matrix to infinity, getting us that

$$\sum_{n=0}^{\infty} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{4} \end{bmatrix}^n = \begin{bmatrix} 2 & 0 \\ 0 & \frac{4}{5} \end{bmatrix}$$

Now multiplying all 3 matrices together and by 5, we get that the final answer comes out to

$$\begin{bmatrix} 6 & 4 \\ 2 & 8 \end{bmatrix}$$

Note that one can also use the geometric sum formula, $\frac{a}{1-r}$, giving us $(I - M)^{-1}$, which would result in the same answer.

13. (B) 1375 The adjacency matrix for the given directed graph is $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$. Multiplying this matrix

with M^4 , we get that $M^5 = \begin{bmatrix} 18 & 55 & 67 \\ 31 & 66 & 91 \\ 30 & 79 & 104 \end{bmatrix}$. We are given that we are at A after 4 moves; thus, we need

to find the number of ways we can get from A to B in 5 moves, which is $M_{12} = 55$. We need to then multiply this number by the numbers of ways we can end at point A after 4 moves; because we start at a random point, this value will be equal to $M_1 + M_2 + M_3 = 25$. Thus, there are $25 \cdot 55 = 1375$

14. (C) 252 If we were to draw some arbitrary z and z^2 with magnitudes 6 and 36 respectively and draw a line connecting x , z , and z^2 , we see that these three points and the origin form a triangle with a cevian corresponding to the line segment formed by connecting z and the origin. Additionally, by $\text{cis}(z)$ notation, we can see that the angle between x and z is equal to that of the angle between z and z^2 . Finally, the magnitudes $|z - x|$ and $|z^2 - z|$ correspond to the lengths of the segments connected z and x and z^2 and z respectively. Thus, through Stewart's Theorem, we get that

$$36 = |z^2 - z||z - x| + 36 \cdot 8 \implies |z^2 - z||z - x| = 252$$

15. (C) $\frac{5}{3}$ Using the basic definition for the Fibonacci sequence, we can change the summation to

$$\sum_{n=3}^{\infty} \frac{2(F_{n+2} - F_n)}{F_n F_{n+2}}.$$

Now, through partial fractions, we can see that this is equal to

$$\sum_{n=3}^{\infty} \frac{2}{F_n} - \frac{2}{F_{n+2}} = \frac{2}{2} - \frac{2}{5} + \frac{2}{3} - \frac{2}{8} + \frac{2}{5} - \frac{2}{13} \dots$$

This sum telescopes to $1 + \frac{2}{3} = \frac{5}{3}$

16. (D) $12\sqrt{3}$ We can manipulate this expression such that AM-GM will work for us; we can see that

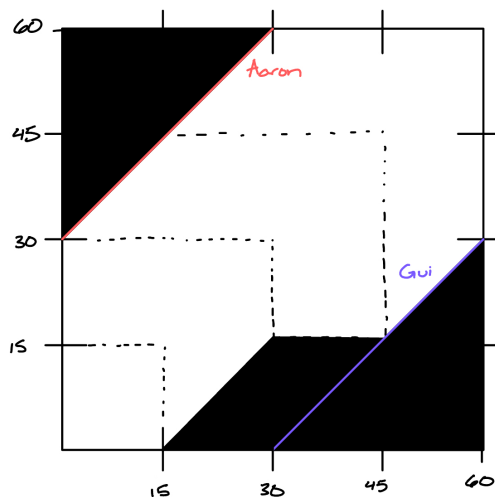
$$\frac{a^2 + ab^2 + 7b^4}{a - b^2} \Rightarrow \frac{(a - b^2)^2 + 3ab^2 + 6b^4}{a - b^2} \Rightarrow (a - b^2) + \frac{3ab^2 + 6b^4}{a - b^2}.$$

We can now substitute $3ab^2 + 6b^4$ using the given; thus, by AM-GM,

$$(a - b^2) + \frac{108}{a - b^2} \geq 2\sqrt{108} \Rightarrow (a - b^2) + \frac{108}{a - b^2} \geq 12\sqrt{3}$$

17. (A) 54 A good thing to note when dealing with floors and ceilings are their differences, that is, for any $x \notin \mathbb{Z}$, $\lceil x \rceil - 1 = \lfloor x \rfloor$. We can use this fact to see that if there are 997 or more positive integers less than or equal to 1000 such that $\log_b(i)$ is not an integer, the inequality will not hold true. Thus, we are finding the sum of the numbers b such that there are at least 4 non-negative integers x such that $b^x \leq 1000$. These numbers are 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10; however, 1 is extraneous; thus, the answer is $2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 54$

18. (B) $\frac{11}{16}$ Because Tiger's arrival is either set or dependent on Gui's arrival, we **do not** have to use 3D Geometry for this question. This also means that Gui and Tiger will always meet up; we can calculate that the probability that Gui and Aaron meet up is $\frac{3}{4}$ through geometric geometry; the only other "bad" case is when Aaron arrives 15 minutes or earlier and Gui arrives after 30 minute or when Gui arrives in between 2:15 PM and 2:30 PM while Aaron arrives 15 minutes earlier. Using basic geometry, we get that any new "bad" areas have an area of $\frac{1}{16}$. Thus, the probability that the 3 of them meet up is $\frac{3}{4} - \frac{1}{16} = \frac{11}{16}$. Below is the diagram for this question.



19. (B) 9 We can easily see that $\det M = P^2(x)$. However, using the given, we can see that $\text{Tr}(M) = R_1(x)R_3(x) + R_2(x)R_1(x) + R_3(x)R_2(x) = \frac{1}{6}P'(x)$. Also using the given, we can see that $(\det M)' = (P^2(x))' = 2P(x)P'(x)$. Plugging all these values in an evaluating, we get that

$$\frac{3(\det M)'}{4\text{Tr}(M)P(x)} \Rightarrow \frac{3 \cdot 2P(x)P'(x)}{4 \cdot \frac{1}{6}P'(x) \cdot P(x)} \Rightarrow 9$$

20. (D) $\pi - \arctan\left(\frac{5}{c}\right)$ After drawing the diagram, we can use Pythagorean Theorem to see that $AP = \sqrt{AB^2 + BP^2}$. Using the given information about eccentricity, we notice that $BP = 4$; thus, $AP = 5$. This means that $\angle AF_1P = \arctan \frac{5}{c}$, meaning that $\angle PF_1F_2 = \pi - \arctan \frac{5}{c}$.